IMPROVED KISSING NUMBERS IN TWENTY-EIGHT THROUGH THIRTY-ONE DIMENSIONS

ABSTRACT. We prove that the kissing numbers in 28, 29, 30 and 31 dimensions are at least 205172, 210473, 222072 and 241077. The best previously published lower bounds were 204522 and 209497, due to Takhanov and Yun, 220440, due to Ma et al., and 238354, due to Takhanov and Yun. All four constructions start from the Leech lifting of Ma et al., which replaces some minimal vectors of the Leech lattice by two or three points with short components in the extra coordinates and adds points in the extra coordinates alone. Ma et al. lift sets of 496 Leech vectors. We lift sets of 528, built from cosets of a five-dimensional subcode of the binary Golay code that contains no octad. We also add points built from Leech lattice vectors of squared norm 64 and 48 together with a direction in the extra coordinates, and in dimensions 28, 29 and 31 we remove some of the points of Ma et al. in the extra coordinates to make room. In dimension 29 we further reweight four hole points and add a fifth pole and two points with rational heads. Each configuration is checked on all pairs by an exact program, with no symmetry reduction and no floating point.

1. INTRODUCTION

The kissing problem in $\mathbb{R}^n$ asks how many unit spheres can touch a central unit sphere without overlapping [2, Ch. 1, §2.2]. In terms of the points of tangency, it asks for the largest set X of unit vectors in $\mathbb{R}^n$ with $\langle x, y\rangle \le 1/2$ for all distinct $x, y \in X$. We write τ_n for this maximum, as in [2, Ch. 13], and call such a set X a *kissing configuration*. In dimension 24 the 196560 minimal vectors of the Leech lattice attain τ_{24} [5, 7].

For $25 \le n \le 31$ the best constructions start from this configuration. Write $n = 24 + d$. We call the first 24 coordinates the *head* and the last d the *tail*. Ma et al. [6] state a general construction, which they attribute in this form to Cohn, Jiao, Kumar and Torquato, as improved by Kallal, Kan and Wang. We call its output a *Leech lift*. It removes some pairwise disjoint sets of Leech minimal vectors, puts each removed vector back as two or three points with short tails, and adds points with zero head. We call the points that keep a Leech minimal vector unchanged the *body*, the points built from a removed vector the *lifts*, and the added points with zero head the *poles*. Ma et al. obtained 204520, 209496, 220440 and 238350 in dimensions 28 to 31. Takhanov and Yun [8] rotate the block of lifts and poles while keeping the body fixed, which makes room for two more points in dimension 28 and one in dimension 29. In dimension 31 they rotate only the tails of the lifts, which makes room for four more points. This gives 204522, 209497 and 238354. Kravatskiy's public repository [3] lists larger values, 204896, 209968, 221012 and 238662, which we have not checked.

Theorem 1.1. *We have* $\tau_{28} \ge 205172$, $\tau_{29} \ge 210473$, $\tau_{30} \ge 222072$ *and* $\tau_{31} \ge 241077$.

Table 1 compares these values with the best published lower bounds, the repository values and the known upper bounds. The gains over the best published values are 650, 976, 1632 and 2723, and over the repository values 276, 505, 1060 and 2415.

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TABLE 1. Kissing number bounds in dimensions 28 to 31, as of October 2026 [1]. The previous lower bounds are the best published ones. The repository values are those listed in [3]; they are not published in a paper and we have not checked them. The upper bounds are the floors of numerical semidefinite programming values [4] and are not interval-certified.

dimension	previous lower bound	repository	new lower bound	upper bound
28	204522 [8]	204896	**205172**	752292 [4]
29	209497 [8]	209968	**210473**	1075991 [4]
30	220440 [6]	221012	**222072**	1537707 [4]
31	238354 [8]	238662	**241077**	2213487 [4]

TABLE 2. The parts of the four configurations.

dimension	body	lifts	poles	layer	holes	rational	total
28	192336	12672	4	160	0	0	205172
29	189168	21120	5	144	34	2	210473
30	183888	38016	72	24	72	0	222072
31	174384	66528	125	40	0	0	241077

The size of a Leech lift in dimension $24 + d$ is

$$196560 + \sum_i (|T_i| - 1)\,|S_i| + |P|,$$

where the S_i are the sets of removed Leech vectors, which we call *head sets*, the T_i are the sets of tails attached to them, and P is the set of poles (Section 2). The tails of [6] in dimensions 28 to 31 are the roots of root systems of type D_4, D_5, E_6 and E_7, and they already make $\sum_i(|T_i| - 1)$ as large as the tails allow. We keep them. The first change is in the head sets. A head set must have pairwise inner products at most $1/4$ after normalization. Ma et al. used head sets of 496 vectors; we use 528. Adding one vector to every head set adds $\sum_i(|T_i| - 1)$ points, namely 16, 26, 48 and 84 in dimensions 28 to 31. So the larger head sets alone give $32 \cdot 16 = 512$, 832, 1536 and 2688 points more than Ma et al.

The second change adds points that are neither lifts nor poles. A *layer point* is a unit vector proportional to $(v, 8w)$, where v is a vector of the Leech lattice of squared norm 64 supported on four coordinates and w is a unit vector in the tail. *Hole points* are built from Leech lattice vectors of squared norm 48. In dimension 30 they have the shape of a lift, and in dimension 29 they are weighted sign vectors in the tail. The final dimension-29 configuration also uses two rational-head points and four reweighted holes, described in Section 8. The original layer and hole shapes are compatible with the body because two distinct vectors of the Leech lattice are at squared distance at least 32. Against a layer vector $v = 4\sigma$ on a 4-set R of coordinates, a head h has $\langle h, v\rangle \le 16$ when h lies in a head set on an octad through R whose matching has exactly one point of R in each pair (Lemma 4.4). This is what allows layer points next to the lifts. In dimensions 28, 29 and 31 we remove some poles to make room. With the changes in the poles, the second change gives 140, 145, 96 and 39 more points. Table 2 gives the parts of each configuration: 160 layer points in dimension 28, 144 layer points, 34 hole points and two rational-head points in dimension 29, 24 layer points and 72 hole points in dimension 30, and 40 layer points in dimension 31.

The paper is organized as follows. Section 2 states the Leech lift and proves its pair conditions. Section 3 describes the tails and the poles of [6]. Section 4 constructs head sets of 528 vectors. Section 5 says which head sets each dimension uses. Section 6 defines layer and hole points and computes their inner products with the other points. Sections 7 to 10 give the four configurations. Section 11 describes the exact checks.

Some steps of the proofs are finite facts about explicit sets that we checked by computer rather than by hand. Examples are the existence of the 512-point parts of the head sets, the disjointness of the head sets, and the largest inner product between a given head set and a given layer vector. We say in each place which facts these are.

2. The Leech lift

Let $\Lambda \subset \mathbb{R}^{24}$ be the Leech lattice, scaled so that its minimal vectors have squared norm 32 and integer coordinates [2, Ch. 4, §11], and let $\Lambda_{\min}$ be the set of its 196560 minimal vectors. At this scale all inner products in Λ are multiples of 8, and two distinct vectors of Λ are at squared distance at least 32. In particular, inner products of distinct vectors of $\Lambda_{\min}$ lie in $\{0, \pm 8, \pm 16, -32\}$. The vectors of Λ of squared norm 48 and 64 are those of type 3 and type 4 in the terminology of [2, Ch. 10].

Throughout we test compatibility exactly. For nonzero x, y with entries in $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ the angle between them is at least 60° if and only if

$$\langle x, y\rangle \le 0 \quad \text{or} \quad 4\langle x, y\rangle^2 \le |x|^2|y|^2. \tag{1}$$

We call such a pair *compatible*. If every pair from a finite set of nonzero vectors is compatible, then the vectors normalized to unit length form a kissing configuration, and distinct vectors give distinct points because (1) excludes cosine 1.

A *head set* is a subset $S \subseteq \Lambda_{\min}$ with $\langle u, u'\rangle \le 8$ for all distinct $u, u' \in S$, that is, with pairwise cosines at most $1/4$.

Write $n = 24 + d$ and $\alpha = \sqrt{2/3}$, $\beta = \sqrt{1/3}$, so $\alpha^2 + \beta^2 = 1$. For $u \in \Lambda_{\min}$ let $\hat{u} = u/\sqrt{32}$. The data of a Leech lift are:

- pairwise disjoint head sets $S_1, \dots, S_k$;
- sets $T_1, \dots, T_k$ of unit vectors in $\mathbb{R}^d$, the *tail sets*, with $\langle t, t'\rangle \le -\frac{1}{2}$ for distinct t, t' in one tail set and $\langle t, t'\rangle \le \frac{1}{2}$ for t, t' in different tail sets;
- a set P of unit vectors in $\mathbb{R}^d$, the *poles*, with $\langle p, p'\rangle \le \frac{1}{2}$ for distinct poles and $\langle t, p\rangle \le \frac{\sqrt{3}}{2}$ for every tail $t \in \bigcup T_i$ and pole p.

The configuration is $X = B \cup L \cup Z$ with

$$\begin{aligned} B &= \{(\hat{u}, 0) : u \in \Lambda_{\min} \setminus \textstyle\bigcup_i S_i\} && \text{(body)},\\ L &= \{(\alpha\hat{u}, \beta t) : u \in S_i,\ t \in T_i,\ 1 \le i \le k\} && \text{(lifts)},\\ Z &= \{(0, p) : p \in P\} && \text{(poles)}. \end{aligned}$$

This is the lifting construction of [6], written with our normalization, together with the poles that [6] add after it.

Proposition 2.1. *X is a kissing configuration in $\mathbb{R}^{24+d}$, and*

$$|X| = 196560 + \sum_{i=1}^{k} (|T_i| - 1)\,|S_i| + |P|. \tag{2}$$

Proof. All points of X are unit vectors. We go through the pairs of distinct points.

Body, body. The inner product is $\langle u, u'\rangle/32 \le 16/32 = \frac{1}{2}$.

Body, lift. The inner product is $\alpha\langle u, u'\rangle/32$ with $u \neq u'$, because the body and the head sets are disjoint. It is at most $\alpha/2 < \frac{1}{2}$.

Lift, lift. Take $(\alpha\hat{u}, \beta t)$ with $u \in S_i$, $t \in T_i$ and $(\alpha\hat{u}', \beta t')$ with $u' \in S_j$, $t' \in T_j$. The inner product is

$$\tfrac{2}{3} \cdot \tfrac{\langle u,u'\rangle}{32} + \tfrac{1}{3}\langle t, t'\rangle.$$

If $i = j$ and $u = u'$, then $t \neq t'$ lie in one tail set, and the value is at most $\frac{2}{3} - \frac{1}{3} \cdot \frac{1}{2} = \frac{1}{2}$. If $i = j$ and $u \neq u'$, then $\langle u, u'\rangle \leq 8$ because S_i is a head set, and the value is at most $\frac{2}{3} \cdot \frac{1}{4} + \frac{1}{3} = \frac{1}{2}$. If $i \neq j$, then $u \neq u'$ because $S_i \cap S_j = \emptyset$, so $\langle u, u'\rangle \leq 16$, and $\langle t, t'\rangle \leq \frac{1}{2}$; the value is at most $\frac{2}{3} \cdot \frac{1}{2} + \frac{1}{3} \cdot \frac{1}{2} = \frac{1}{2}$.

Lift, pole. The inner product is $\beta\langle t, p\rangle \leq \frac{1}{\sqrt{3}} \cdot \frac{\sqrt{3}}{2} = \frac{1}{2}$.

Pole, pole and *body, pole.* The inner products are $\langle p, p'\rangle \leq \frac{1}{2}$ and 0.

So all cosines are at most $\frac{1}{2}$, which also shows that the points are distinct. The count is $(196560 - \sum |S_i|) + \sum |S_i||T_i| + |P|$. □

The proof never uses which head set is attached to which tail set. So any assignment of the head sets to the tail sets gives a kissing configuration. Sections 7 to 10 use this freedom.

Lemma 2.2. *(i) Every tail set has at most 3 tails. (ii) If the tails form a set $\mathcal{T}$, then $\sum_i(|T_i| - 1) \leq \lfloor 2|\mathcal{T}|/3\rfloor \leq \lfloor 2\tau_d/3\rfloor$.*

Proof. (i) For unit vectors $t_1, \ldots, t_m$ with pairwise inner products at most $-\frac{1}{2}$ we have $0 \leq |\sum t_j|^2 \leq m - m(m-1)/2$, so $m \leq 3$. (ii) All tails together form a kissing configuration in $\mathbb{R}^d$, so $|\mathcal{T}| \leq \tau_d$, and $|T_i| - 1 \leq \frac{2}{3}|T_i|$ by (i). □

A tail set with three tails is a *triangle*: $\{a, b, -a - b\}$ with $\langle a, b\rangle = -\frac{1}{2}$. A tail set with two tails at inner product -1 is an *antipodal pair*. Each triangle contributes $2|S_i|$ to (2) and each pair $|S_i|$.

3. Tails and poles

A tail is a unit vector; we call $\sqrt{2}\,t$ the *root* of a tail t. The coordinate lists of Section 11 use other scalings, which we state where we use them. We use the tails of [6], as given in their published configurations, and in dimensions 30 and 31 also their poles. We describe them by the following finite facts, which we checked in exact arithmetic on the coordinate lists. In each dimension $d \in \{4, 5, 6, 7\}$:

(1) The roots are closed under the reflections they define and have pairwise inner products in $\{0, \pm 1, -2\}$. They span $\mathbb{R}^d$, and their number is 24, 40, 72, 126. So they are the roots of a root system of type D_4, D_5, E_6, E_7; no other simply laced root system of rank d has this many roots.

(2) The poles of [6], scaled to squared norm 2, have the same properties as in (1), so they form a second root system of the same type.

(3) The cosine between a tail and a pole is at most $\frac{\sqrt{3}}{2}$.

Since distinct roots of a simply laced root system have cosines in $\{0, \pm\frac{1}{2}, -1\}$, the tail set conditions of Section 2 hold for any partition of the tails into triangles $\{a, b, -a - b\}$ and antipodal pairs, and the pole–pole condition holds automatically. Facts (1) and (2) identify the two root systems; the only inner products that are not forced by the root system structure are those of fact (3), a check of 24^2, 40^2, 72^2 and 126^2 inner products.

In every case $\sum_i(|T_i| - 1)$ equals 16, 26, 48 and 84, which is $\lfloor 2|\mathcal{T}|/3\rfloor$, the bound of Lemma 2.2(ii). So, for these tails, the only room left in (2) is in the sizes $|S_i|$ and in the

TABLE 3. Tails, tail sets and the poles of [6]. The last column is the largest cosine between a tail and a pole.

dimension	d	tails	tail sets	poles	tail–pole
28	4	D_4, 24 roots	8 triangles	D_4, 24	$\sqrt{3}/2$
29	5	D_5, 40 roots	12 triangles, 2 pairs	D_5, 40	$(2+\sqrt{2})/4 \approx 0.8536$
30	6	E_6, 72 roots	24 triangles	E_6, 72	$\sqrt{3}/2$
31	7	E_7, 126 roots	42 triangles	E_7, 126	$(2+\sqrt{2})/4$

poles. The configurations in dimensions 30 and 31 keep the poles of [6] (minus one in dimension 31). In dimensions 28 and 29 we use fewer poles, and the largest tail–pole cosine in Table 3 for dimension 29 does not matter for our configuration.

In dimension 29 the tails, at squared norm 16, are the vectors $2\sqrt{2}(\pm e_i \pm e_j)$ in the coordinates $e_1, \dots, e_5$ of our coordinate lists. The 14 tail sets of [6], written without the factor $2\sqrt{2}$, are

$$\{-e_1-e_2,\ e_2+e_3,\ e_1-e_3\},\quad \{-e_1-e_4,\ e_2+e_4,\ e_1-e_2\},$$
$$\{-e_1+e_5,\ -e_2-e_5,\ e_1+e_2\},\quad \{-e_1+e_4,\ -e_4+e_5,\ e_1-e_5\},$$
$$\{-e_1+e_3,\ -e_3-e_4,\ e_1+e_4\},\quad \{-e_1+e_2,\ -e_2+e_4,\ e_1-e_4\},$$
$$\{-e_2-e_3,\ e_3+e_4,\ e_2-e_4\},\quad \{-e_2-e_4,\ e_4+e_5,\ e_2-e_5\},$$
$$\{-e_2+e_5,\ e_3-e_5,\ e_2-e_3\},\quad \{-e_2+e_3,\ -e_3-e_5,\ e_2+e_5\},$$
$$\{-e_3+e_5,\ e_4-e_5,\ e_3-e_4\},\quad \{-e_3+e_4,\ -e_4-e_5,\ e_3+e_5\},$$
$$\{\pm(e_1+e_3)\},\quad \{\pm(e_1+e_5)\}.$$

In the other dimensions the coordinates are less pleasant, and the partition of the tails into triangles is part of the coordinate lists that accompany the paper. The proofs use the partition only through Proposition 2.1 and Lemma 7.1, which hold for every partition into triangles, and through the computed facts (C5), (C8), (C9) and (C12) of Sections 9 and 10.

4. HEAD SETS OF 528 VECTORS

4.1. **The Golay code in the shell.** Index the coordinates of $\mathbb{R}^{24}$ by $\{0, 1, \dots, 23\}$, as in our coordinate lists, and identify binary words of length 24 with subsets of $\{0, \dots, 23\}$. The vectors of $\Lambda_{\min}$ have three shapes [2, Ch. 4, §11]: 1104 of shape $(\pm 4, \pm 4, 0^{22})$, 97152 of shape $(\pm 2^8, 0^{16})$ and 98304 of shape $(\mp 3, \pm 1^{23})$. For a position i and a word a, let $v(i,a)$ be the vector with entry -3 at i and 1 elsewhere, with the signs of the coordinates in a changed. In the description of Λ in [2, Ch. 4, §11], a vector with odd entries lies in Λ only if its coordinates congruent to 3 modulo 4 form a word of a fixed copy G of the extended binary Golay code. For $v(i,a)$ these coordinates are exactly a. So the vectors of shape $(\mp 3, \pm 1^{23})$ are exactly the $24 \cdot 4096$ vectors $v(i,a)$ with $a \in G$, and the supports of the vectors of shape $(\pm 2^8, 0^{16})$ are the 759 octads of the same G. For $c \in G$ let S_c change the signs of the coordinates in c. It maps $\Lambda_{\min}$ to itself and sends $v(i,a)$ to $v(i, a+c)$.

Lemma 4.1 (conflict rule)**.** *Let $a, b \in G$ and $c = a + b$. Then $\langle v(i,a), v(j,b)\rangle = 16$ in exactly these cases: $i = j$ and c is an octad not containing i; $i \neq j$ and $c = 0$; $i \neq j$ and c is an octad containing i and j. In all other cases with $v(i,a) \neq v(j,b)$ the inner product is at most 8.*

Proof. Let $w = \mathrm{wt}(c)$ and $\epsilon_k = -1$ if $k \in c$, $\epsilon_k = 1$ otherwise, so that $\sum_k \epsilon_k = 24 - 2w$. The two vectors agree in sign exactly off c. If $i = j$, the inner product is $\sum_{k \neq i} \epsilon_k + 9\epsilon_i = 24 - 2w + 8\epsilon_i$.

If $i \neq j$, it is $\sum_{k\neq i,j} \epsilon_k - 3\epsilon_i - 3\epsilon_j = 24 - 2w - 4\epsilon_i - 4\epsilon_j$. The weights in G are $0, 8, 12, 16, 24$, and the values follow. □

4.2. **The 512-point part.** Let $K \leq G$ be a subcode that contains no octad. Its nonzero words have weights in $\{12, 16, 24\}$, so its minimum distance is at least 12. If $c \in K$ is nonzero, then c is not an octad, so by Lemma 4.1 the 32 vectors $v(i, a + c)$, $c \in K$, have pairwise inner products at most 8. Every octad-free subcode has dimension at most 5: a binary $[24, 6, 12]$ code would violate the Griesmer bound, since $12 + 6 + 3 + 2 + 1 + 1 = 25 > 24$.

We use five such subcodes $K_1, \ldots, K_5$. Each has dimension 5 and weight distribution $0^1\, 12^{28}\, 16^3$. The complements of the three words of weight 16 in K_f are three pairwise disjoint octads, a *trio*. Table 4 lists a basis of each subcode and its trio. These are finite computations. The subcodes K_4 and K_5 are used only in dimension 29.

Fix $K = K_f$ and an octad O of its trio. For each position $i \notin O$ choose a coset $\gamma_i \in G/K$, and let

$$A = \{v(i, a) : i \notin O,\ a \in \gamma_i\},$$

a set of $16 \cdot 32 = 512$ vectors of shape $(\mp 3, \pm 1^{23})$. We use only the 16 positions outside O because the positions of O are kept for the 16-point part below: Lemma 4.2 needs the entries -3 of A to lie outside O. By Lemma 4.1, A is a head set if and only if for all positions $i \neq j$ outside O, the coset $\gamma_i + \gamma_j$ contains neither 0 nor an octad through i and j; the case $i = j$ is handled by octad-freeness. In particular the γ_i are distinct. If (γ_i) satisfies this condition, so does $(\gamma_i + t)$ for every $t \in G/K$. These 128 translates are pairwise disjoint, because they use different cosets at every position. We call a head set of this form a *512-point part on* (K, O).

For each pair (K_f, O) that we use, an exact search found such an assignment (γ_i). For K_1, an exhaustive search over all 759 octads found such assignments only on the three octads of its trio, and none on 400 random 8-sets that are not octads. We did not prove this in general, and the construction does not depend on it.

4.3. **The 16-point part.**

Lemma 4.2. *Let O be an octad and let A be a set of vectors of shape $(\mp 3, \pm 1^{23})$ whose entries ∓ 3 all lie outside O. Let M be a perfect matching of O (four disjoint pairs covering O), and let $F(M)$ be the 16 vectors $(\pm 4, \pm 4, 0^{22})$ supported on the pairs of M, with all four sign choices on each pair. Then every vector of $F(M)$ has inner product in $\{-8, 0, 8\}$ with every vector of A, and the vectors of $F(M)$ have pairwise inner products in $\{0, -32\}$. Conversely, every set of vectors of shape $(\pm 4, \pm 4, 0^{22})$ supported in O with pairwise inner products at most 8 has at most 16 elements.*

Proof. The vectors of A have entries ± 1 on O, so a vector $4(\pm e_p \pm e_q)$ with $p, q \in O$ meets them in $4(\pm 1 \pm 1)$. Two vectors of $F(M)$ on different pairs have disjoint supports, and two on the same pair have inner product 0 or -32. For the converse, two such vectors whose supports share exactly one point p and agree in sign at p have inner product 16. At p there are two signs, and for each sign the vectors through p with that sign at p must have the same partner, so at most four vectors pass through p. Counting incidences, there are at most $8 \cdot 4/2 = 16$ vectors. □

Theorem 4.3. *Let A be a 512-point part on (K, O) and M a perfect matching of O. Then $A \cup F(M)$ is a head set of 528 vectors.*

Proof. A is a head set by the choice of (γ_i), $F(M)$ is one by Lemma 4.2, and the cross products are at most 8 by the same lemma. □

The head sets of Ma et al. have 496 vectors [6]. We do not know the largest head set.

4.4. **Disjoint head sets.** Two head sets $A \cup F(M)$ and $A' \cup F(M')$ are disjoint if and only if $A \cap A' = \emptyset$ and M and M' have no pair in common, because the two parts have different shapes and $F(M) \cap F(M')$ consists of the vectors on common pairs.

Matchings. An octad has $\binom{8}{2} = 28$ pairs and each matching uses 4, so at most 7 of our head sets lie on one octad. The three octads of one trio are disjoint, so one subcode gives at most $3 \cdot 7 = 21$ head sets. Octads from different trios share 0, 2 or 4 points, and shared pairs can be used only once. For the trios of K_1 and K_2 the six octads contain 140 distinct pairs, which allows at most 35 matchings.

512-point parts. Two translates of one part are disjoint. For parts of one subcode on different octads, and for parts of different subcodes, whether they meet is a condition on the translation labels. For K_f and K_g it depends only on the images of the labels in $G/(K_f + K_g)$, because two cosets of K_f and K_g at a common position meet exactly when their representatives agree modulo $K_f + K_g$. For K_1, K_2, K_3 we have $|K_f \cap K_g| = 4$ for $f \neq g$, so each $K_f + K_g$ has dimension 8, and a computation shows that $K_f + K_g = K_1 + K_2 + K_3$. So all cross conditions live in one quotient of order 16, and a small exhaustive search finds labels that satisfy all of them.

We do not need the details of these searches for the proofs: in each dimension the head sets are listed explicitly in the coordinate lists, and we checked by computer that they are pairwise disjoint (Section 11).

4.5. **Head sets and a tetrad.** A *tetrad* is a set of four coordinates. The five octads that contain a tetrad R partition the other twenty coordinates into five tetrads, and these six tetrads form a *sextet* [2, Ch. 10]. The layer points of Section 6 involve the vectors 4σ supported on a tetrad R, with $\sigma \in \{\pm 1\}^R$. All 16 of them lie in Λ, since 4σ is the sum of two vectors of shape $(\pm 4, \pm 4, 0^{22})$ with disjoint supports. Each has squared norm 64. We need to know how they meet head sets.

Lemma 4.4. *Let R be a tetrad contained in an octad O, let A be a set of vectors of shape $(\mp 3, \pm 1^{23})$ whose entries ∓ 3 lie outside O, and let M be a perfect matching of O in which every pair has exactly one point in R. Then $|\langle h, 4\sigma\rangle| \leq 16$ for every $h \in A \cup F(M)$ and every $\sigma \in \{\pm 1\}^R$. If instead M has a pair inside R, then $\langle h, 4\sigma\rangle = 32$ for some $h \in F(M)$ and some σ.*

Proof. A vector of A has entries ± 1 on R, so it meets 4σ in $4\sum_{k\in R} \pm 1$, which has absolute value at most 16. A vector $4(\pm e_p \pm e_q)$ with exactly one of p, q in R meets 4σ in ± 16. If $p, q \in R$, choosing the signs to agree gives 32. □

We say a matching M of $O \supseteq R$ is *R-bipartite* if every pair of M has exactly one point in R. The R-bipartite pairs of O form a complete bipartite graph $K_{4,4}$, which has 16 edges, so at most four pairwise pair-disjoint R-bipartite matchings exist on one octad. Four exist: they form a 1-factorization of $K_{4,4}$.

Lemma 4.5 (sextet)**.** *Every tetrad R lies in exactly five octads, and any two of them meet exactly in R.*

Proof. Every 5-set lies in exactly one octad, because the octads form a Steiner system $S(5, 8, 24)$ [2, Ch. 10]. So each of the 20 points outside R lies in exactly one octad through R, and each such octad contains 4 of them. □

So R-bipartite matchings on different octads through R never share a pair, since a shared pair would lie in the intersection R. At most $5 \cdot 4 = 20$ head sets can have R-bipartite matchings.

TABLE 4. The five subcodes. Each row gives a basis: two words of weight 16 and three of weight 12. The trio consists of the complements of the weight-16 words of K_f.

K_1	$\{0,1,3,5,6,7,8,10,12,13,14,15,16,18,19,20\}$, $\{2,4,5,6,8,9,11,12,13,15,16,17,18,21,22,23\}$, $\{0,1,2,3,5,8,11,13,15,17,19,22\}$, $\{0,1,2,4,6,8,9,10,13,18,20,22\}$, $\{0,2,3,4,6,10,11,12,13,14,15,21\}$
trio	$O_{11} = \{0,1,3,7,10,14,19,20\}$, $O_{12} = \{2,4,9,11,17,21,22,23\}$, $O_{13} = \{5,6,8,12,13,15,16,18\}$
K_2	$\{0,1,3,4,6,7,10,12,14,16,17,18,19,20,21,22\}$, $\{2,3,4,5,7,8,9,11,12,13,14,15,16,17,20,23\}$, $\{0,1,2,3,4,5,7,10,12,13,15,19\}$, $\{0,1,2,3,9,11,12,13,14,17,21,22\}$, $\{0,2,3,4,5,6,9,10,17,20,22,23\}$
trio	$O_{21} = \{0,1,6,10,18,19,21,22\}$, $O_{22} = \{2,5,8,9,11,13,15,23\}$, $O_{23} = \{3,4,7,12,14,16,17,20\}$
K_3	$\{0,2,4,5,7,8,9,10,11,13,14,18,19,20,21,22\}$, $\{1,2,3,4,6,7,10,11,12,14,15,16,17,20,21,23\}$, $\{0,1,2,3,4,5,8,9,11,12,14,15\}$, $\{0,1,2,3,4,8,13,17,20,21,22,23\}$, $\{0,1,2,5,6,7,14,15,18,21,22,23\}$
trio	$O_{31} = \{0,5,8,9,13,18,19,22\}$, $O_{32} = \{1,3,6,12,15,16,17,23\}$, $O_{33} = \{2,4,7,10,11,14,20,21\}$
K_4	$\{0,1,2,3,4,5,6,9,10,12,13,16,17,18,19,20\}$, $\{0,1,4,7,8,10,11,12,14,15,16,17,19,21,22,23\}$, $\{5,6,7,9,10,11,12,14,15,16,18,19\}$, $\{1,2,3,6,7,8,9,10,15,16,17,21\}$, $\{1,2,4,6,7,8,11,12,13,16,18,22\}$
trio	$O_{41} = \{0,1,4,10,12,16,17,19\}$, $O_{42} = \{2,3,5,6,9,13,18,20\}$, $O_{43} = \{7,8,11,14,15,21,22,23\}$
K_5	$\{0,1,4,5,6,9,10,11,12,15,16,17,18,19,21,22\}$, $\{0,1,2,3,5,7,8,9,10,11,13,14,15,19,20,23\}$, $\{0,1,4,7,8,9,11,12,13,14,16,17\}$, $\{2,4,6,7,9,10,11,13,16,18,19,20\}$, $\{0,2,3,4,6,10,11,12,13,14,15,21\}$
trio	$O_{51} = \{0,1,5,9,10,11,15,19\}$, $O_{52} = \{4,6,12,16,17,18,21,22\}$, $O_{53} = \{2,3,7,8,13,14,20,23\}$

TABLE 5. Number of head sets on each octad of Table 4. In dimension 29 there are also 4 head sets on O_{41} and 2 on O_{51}.

dimension	O_{11}	O_{12}	O_{13}	O_{21}	O_{22}	O_{23}	O_{31}	O_{32}	O_{33}	total
28	4	–	–	4	–	–	–	–	–	8
29	4	–	–	4	–	–	–	–	–	14
30	–	6	5	6	–	7	–	–	–	24
31	6	7	6	4	3	3	4	5	4	42

5. The head sets in each dimension

In each dimension the configuration contains a Leech lift built from the tails of Section 3 and one head set of Theorem 4.3 on each tail set: $k = 8$, 14, 24, 42 pairwise disjoint head sets of 528 vectors. By (2) its body and lifts together have $196560 + 528 \sum_i (|T_i| - 1)$ points, that is $196560 + 528 \cdot 16$, $+528 \cdot 26$, $+528 \cdot 48$ and $+528 \cdot 84$. Proposition 2.1 applies to any assignment of disjoint head sets to tail sets, so the assignment is free for the later sections. Table 5 lists how many head sets lie on each octad. Their matchings are pairwise pair-disjoint, and their 512-point parts are pairwise disjoint by computer check.

Dimensions 28 and 29. All head sets lie on octads through $S = \{0,1,10,19\}$, so that they are compatible with the layer points on S (Sections 7 and 8).

Dimension 30. Twenty-four head sets are more than the 21 that one subcode can give, so we use K_1 and K_2.

Dimension 31. Forty-two head sets are more than the 35 that the trios of K_1 and K_2 can give, so we use all three of K_1, K_2, K_3. The 42 matchings use 168 distinct coordinate pairs out of the 177 pairs that lie in the nine octads.

6. Layer points and hole points

We now add points to a Leech lift, and sometimes remove poles. In this section we define the new points and compute their inner products with the old ones. We keep the notation of Section 2: body points $(\hat{b}, 0)$, lifts $(\alpha\hat{h}, \beta t)$ with h in a head set and t a tail of its tail set, and poles $(0, p)$ with p a unit vector. We use the following consequence of the minimum distance of Λ: if $x, y \in \Lambda$ are distinct, then

$$\langle x, y\rangle \le \tfrac{1}{2}\big(|x|^2 + |y|^2 - 32\big). \tag{3}$$

For $|x|^2 = 64$ and $y \in \Lambda_{\min}$ this gives $\langle x, y\rangle \le 32$; for $|x|^2 = 48$ and $y \in \Lambda_{\min}$ it gives $\langle x, y\rangle \le 24$; for $|x|^2 = 48$ and $|y|^2 = 64$ it gives $\langle x, y\rangle \le 40$.

6.1. **Layer points.** Let $v \in \Lambda$ with $|v|^2 = 64$ and let $w \in \mathbb{R}^d$ be a unit vector. The *layer point* with data (v, w) is the unit vector

$$Y(v, w) = \Big(\frac{v}{8\sqrt{2}},\ \frac{w}{\sqrt{2}}\Big),$$

which is proportional to $(v, 8w)$. In all our layer points, $v = 4\sigma$ is supported on a tetrad R as in Section 4.5. The parity of σ is the parity of the number of its minus signs. For two sign vectors σ, σ' on R that differ in j places, $\langle 4\sigma, 4\sigma'\rangle = 16(4 - 2j)$. So two distinct sign vectors of the same parity give $\langle v, v'\rangle \in \{0, -64\}$, and two of different parity give ± 32.

Lemma 6.1. *For a layer point $Y = Y(v, w)$:*

(1) $\langle Y, (\hat{b}, 0)\rangle = \langle v, b\rangle/64 \le \frac{1}{2}$ *for every* $b \in \Lambda_{\min}$*;*

(2) $\langle Y, (\alpha\hat{h}, \beta t)\rangle = \dfrac{\langle h, v\rangle}{32\sqrt{6}} + \dfrac{\langle t, w\rangle}{\sqrt{6}}$*;*

(3) $\langle Y, (0, p)\rangle = \langle w, p\rangle/\sqrt{2}$*;*

(4) $\langle Y(v, w), Y(v', w')\rangle = \dfrac{\langle v, v'\rangle}{128} + \dfrac{\langle w, w'\rangle}{2}$.

In particular, the lift product in (2) is at most $\frac{1}{2}$ in each of these cases: $\langle h, v\rangle \le 16$ and $\langle t, w\rangle \le \frac{1}{\sqrt{2}}$; $\langle h, v\rangle \le 24$ and $\langle t, w\rangle \le \frac{2\sqrt{6}-3}{4} \approx 0.4747$; $\langle t, w\rangle \le \frac{\sqrt{6}-2}{2} \approx 0.2247$. The pole product in (3) is at most $\frac{1}{2}$ when $\langle w, p\rangle \le \frac{1}{\sqrt{2}}$.

Proof. The four formulas are direct computations. The bound in (1) is (3). In (2) we always have $\langle h, v\rangle \le 32$ by (3), and $\frac{\langle h,v\rangle}{32} + \langle t, w\rangle \le \frac{\sqrt{6}}{2}$ is the condition for the value to be at most $\frac{1}{2}$; the three cases are $\frac{1}{2} + \frac{1}{\sqrt{2}} < 1.2247$, $\frac{3}{4} + 0.4747 < 1.2247$ and $1 + 0.2247 \le 1.2247$. □

6.2. **Hole points.** Let $u \in \Lambda$ with $|u|^2 = 48$. Two kinds of points built from u appear below. In dimension 30 we use

$$x(u, y) = \Big(\alpha\frac{u}{\sqrt{48}},\ \beta y\Big)$$

with $y \in \mathbb{R}^6$ a unit vector, the same proportions as a lift. In dimension 29 we use the unit vector proportional to $(p\sqrt{2}\,u,\ qs)$ with $s \in \{\pm 1\}^5$ and integers

$$(p, q) \in \{(1, 4),\ (3, 10),\ (5, 29),\ (1, 14),\ (10, 31)\};$$

its squared norm before normalizing is $N = 96p^2 + 5q^2$, which is 176, 1364, 6605, 1076 and 14405. We call both kinds *hole points*. The name refers to their tail directions, which point at deep holes of the lattice spanned by the roots: the vectors $s/2$ are the half-spin weights of D_5, and the directions y in dimension 30 are those of minuscule weights of E_6. The bounds for hole points are proved in Sections 8 and 9.

7. Dimension 28

7.1. **The frame.** The 24 tails are the unit vectors of a root system of type D_4. Choose an orthonormal frame $e_1, \dots, e_4$ of $\mathbb{R}^4$ in which the tails are $(\pm e_i \pm e_j)/\sqrt{2}$. The 24 unit vectors

$$\Pi_0 = \{\pm e_i\}, \qquad \Pi_+ = \{\tfrac{1}{2}(\pm 1, \pm 1, \pm 1, \pm 1) \text{ with an even number of minus signs}\},$$
$$\Pi_- = \{\tfrac{1}{2}(\pm 1, \pm 1, \pm 1, \pm 1) \text{ with an odd number of minus signs}\}$$

are the directions of the deep holes of the D_4 root lattice. Each of Π_0, Π_+, Π_- is the vertex set of a cross-polytope: two of its vectors have inner product 0 or -1. Vectors of different classes have inner product $\pm\frac{1}{2}$. A tail has inner product 0 or $\pm\frac{1}{\sqrt{2}}$ with each of the 24 vectors. These facts are short computations in the frame.

Lemma 7.1. *Let $c \in \Pi_0$. Of the eight triangles of tails, exactly two are perpendicular to c, and each of the other six has exactly one tail t with $\langle t, c\rangle = \frac{1}{\sqrt{2}}$.*

Proof. For a triangle $\{a, b, -a-b\}$ the three values $\langle a, c\rangle, \langle b, c\rangle, -\langle a, c\rangle - \langle b, c\rangle$ lie in $\{0, \pm\frac{1}{\sqrt{2}}\}$ and sum to 0. So either all are 0, or they are $0, \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}$ in some order. Twelve of the 24 tails are not perpendicular to c, so six triangles are of the second kind and two of the first. □

7.2. **The head sets.** Let $S = \{0, 1, 10, 19\}$. The octads $O_{11} = S \cup \{3, 7, 14, 20\}$ and $O_{21} = S \cup \{6, 18, 21, 22\}$ both contain S. We use four head sets on each: $A_j \cup F(M_j)$ for $j = 1, \dots, 8$, where $A_1, \dots, A_4$ are pairwise disjoint 512-point parts on (K_1, O_{11}) and $A_5, \dots, A_8$ on (K_2, O_{21}), and where $M_1, \dots, M_4$ and $M_5, \dots, M_8$ are the two 1-factorizations of the S-bipartite pairs of O_{11} and O_{21}:

$$M_1 = \{\text{0-14, 19-20, 1-3, 10-7}\},\ M_2 = \{\text{0-20, 19-14, 1-7, 10-3}\},$$
$$M_3 = \{\text{0-3, 19-7, 1-14, 10-20}\},\ M_4 = \{\text{0-7, 19-3, 1-20, 10-14}\},$$
$$M_5 = \{\text{0-21, 1-22, 10-6, 19-18}\},\ M_6 = \{\text{0-22, 1-21, 10-18, 19-6}\},$$
$$M_7 = \{\text{0-6, 1-18, 10-21, 19-22}\},\ M_8 = \{\text{0-18, 1-6, 10-22, 19-21}\}.$$

The matchings are pairwise pair-disjoint by Lemma 4.5. The disjointness of the 512-point parts on different octads is a computer check. By Lemma 4.4, every head h of these eight sets has $|\langle h, 4\sigma\rangle| \le 16$ for all $\sigma \in \{\pm 1\}^S$.

7.3. **The configuration.** Fix two orthogonal vectors $p_1, p_2 \in \Pi_0$ and let $R_1 = \{0, 1, 6, 18\}$ and $R_2 = \{0, 3, 7, 19\}$. The configuration consists of:

- the body and the lifts of the eight head sets above on the eight triangles, with the assignment described below: $192336 + 12672$ points;
- poles at the four vectors of $\Pi_0 \setminus \{\pm p_1, \pm p_2\}$;
- 64 layer points $Y(4\sigma, a)$ with $a \in \Pi_+$ and $\sigma \in \{\pm 1\}^S$ even, and 64 layer points $Y(4\sigma, b)$ with $b \in \Pi_-$ and σ odd;
- 16 layer points $Y(4\tau, \pm p_1)$ with $\tau \in \{\pm 1\}^{R_1}$ even, and 16 layer points $Y(4\tau, \pm p_2)$ with $\tau \in \{\pm 1\}^{R_2}$ odd.

Assignment. For every head h of the eight sets, $\max_\tau \langle h, 4\tau\rangle$ over the even τ on R_1 is 16 except for the sets on M_7 and M_8, where it is 32; over the odd τ on R_2 it is 16 except for the sets on M_3 and M_4, where it is 32. Since $R_1 \subset O_{21}$ and $R_2 \subset O_{11}$, Lemma 4.4 gives these values on the octad that contains the tetrad: M_5, M_6 are R_1-bipartite and M_1, M_2 are R_2-bipartite, while M_7, M_8 have pairs inside R_1 and M_3, M_4 have pairs inside R_2. The value 16 for the sets on the other octad is a computer check. We put the sets on M_7, M_8 on the two triangles perpendicular to p_1, and the sets on M_3, M_4 on the two triangles perpendicular to p_2. No triangle is perpendicular to both, since the tails perpendicular to two orthogonal vectors of Π_0 are four vectors $(\pm e_k \pm e_l)/\sqrt{2}$, which contain no triangle.

Theorem 7.2. *This is a kissing configuration of* $192336 + 12672 + 4 + 128 + 32 = 205172$ *points in* $\mathbb{R}^{28}$.

Proof. Pairs among the body and the lifts are covered by Proposition 2.1. A lift and a pole have inner product $\beta\langle t, c\rangle \le \frac{1}{\sqrt{6}}$, and two poles have inner product 0 or -1. We use Lemma 6.1 for the layer points. Call the 128 points with directions in $\Pi_+ \cup \Pi_-$ the S-layer and the 32 others the R-layer.

Layer, body. At most $\frac{1}{2}$ by Lemma 6.1(1).

Layer, lift. For the S-layer, $\langle h, 4\sigma\rangle \le 16$ for every head and $\langle t, w\rangle \le \frac{1}{\sqrt{2}}$ for every tail, so the value is at most $\frac{16}{32\sqrt{6}} + \frac{1}{\sqrt{12}} \approx 0.4928$. For the R-layer at $\pm p_1$, a triangle not perpendicular to p_1 carries a set with $\langle h, 4\tau\rangle \le 16$, and the same bound holds. A triangle perpendicular to p_1 has $\langle t, w\rangle = 0$ and $\langle h, 4\tau\rangle \le 32$, giving at most $\frac{1}{\sqrt{6}}$. The same holds at $\pm p_2$.

Layer, pole. The poles lie in Π_0. An S-layer direction has inner product $\pm\frac{1}{2}$ with them, which gives $\frac{1}{2\sqrt{2}}$. The R-layer directions $\pm p_1, \pm p_2$ are orthogonal to the four remaining poles.

Layer, layer. By Lemma 6.1(4) the value is $\langle v, v'\rangle/128 + \langle w, w'\rangle/2$. Same direction: $v \ne v'$ have the same parity on the same tetrad, so $\langle v, v'\rangle \le 0$. Opposite or orthogonal directions: $\langle v, v'\rangle \le 64$, and the value is at most $\frac{1}{2}$. Directions in different classes among Π_0, Π_+, Π_- have $\langle w, w'\rangle = \pm\frac{1}{2}$. For two S-layer points in Π_+ and Π_- the parities differ, so $\langle v, v'\rangle = \pm 32$; for an S-layer point and an R-layer point, $|S \cap R_i| = 2$, so $|\langle v, v'\rangle| \le 32$. In both cases the value is at most $\frac{1}{4} + \frac{1}{4}$. An R_1 point and an R_2 point have orthogonal directions and $|R_1 \cap R_2| = 1$, so the value is at most $\frac{16}{128}$.

So all cosines are at most $\frac{1}{2}$. □

8. Dimension 29

8.1. **The frame and the head sets.** The tails, at squared norm 16, are $2\sqrt{2}(\pm e_i \pm e_j)$ in the coordinates $e_1, \dots, e_5$ of Section 3, in 12 triangles and 2 antipodal pairs. Let $S = \{0, 1, 10, 19\}$ as before. By Lemma 4.5, S lies in five octads; four of them are

$$O_{11} = S \cup \{3, 7, 14, 20\}, \quad O_{21} = S \cup \{6, 18, 21, 22\},$$
$$O_{41} = S \cup \{4, 12, 16, 17\}, \quad O_{51} = S \cup \{5, 9, 11, 15\},$$

which lie in the trios of K_1, K_2, K_4 and K_5. We use 4, 4, 4 and 2 head sets $A \cup F(M)$ on them, with pairwise disjoint 512-point parts on (K_f, O) and S-bipartite matchings M. The matchings on one octad are pairwise pair-disjoint, and by Lemma 4.5 so are matchings on different octads. The disjointness of the 512-point parts is a computer check. By Lemma 4.4, $|\langle h, 4\sigma\rangle| \le 16$ for every head h and every $\sigma \in \{\pm 1\}^S$.

8.2. **Layer points.** Let $\mathcal{V} = \{\pm e_1, \dots, \pm e_5\}$ and let $\mathcal{W}$ be the eight vectors $\pm\frac{1}{2}(\eta_1 e_1 + \eta_2 e_2 + \eta_3 e_4 + \eta_4 e_5)$, where η runs over the rows of a 4×4 Hadamard matrix. The 144 layer points are $Y(4\sigma, w)$ with $w \in \mathcal{V}$ and σ even on S (80 points), and with $w \in \mathcal{W}$ and σ odd (64 points). A tail has inner product in $\{0, \pm\frac{1}{\sqrt{2}}\}$ with every $w \in \mathcal{V}$ and in $\{0, \pm\frac{1}{2\sqrt{2}}, \pm\frac{1}{\sqrt{2}}\}$ with every $w \in \mathcal{W}$. Two vectors of $\mathcal{V}$, or two of $\mathcal{W}$, have inner product 0 or -1; a vector of $\mathcal{V}$ and one of $\mathcal{W}$ have inner product 0 or $\pm\frac{1}{2}$.

8.3. **Poles, hole points and rational-head points.** Four poles have the unit tail vectors

$$\tfrac{1}{\sqrt{5}}(-1,1,1,-1,-1), \quad \tfrac{1}{\sqrt{6}}(1,-1,-1,\sqrt{2},1), \quad \tfrac{1}{\sqrt{6}}(-1,1,-\sqrt{2},-1,-1), \quad \tfrac{1}{\sqrt{6}}(1,-1,\sqrt{2},1,1).$$

The fifth is proportional to $(0^{24}; 16, -25, -50, -26, 34)$, of squared norm 5213.

There are 34 hole points, proportional to $(p\sqrt{2}\,u, qs)$, with $|u|^2 = 48$ and $s \in \{\pm 1\}^5$. Six have $(p,q) = (1,4)$, ten have $(3,10)$, eight have $(5,29)$, six have $(1,14)$ and four have $(10,31)$. The vectors s are twelve sign vectors closed under negation; two carry four hole points, six carry three and four carry two. The u are 17 Leech vectors, eleven of shape $(\pm 4, \pm 2^8, 0^{15})$ and six of shape $(\pm 2^{12}, 0^{12})$.

The four holes with $(p,q) = (10,31)$ have $s = (-1,-1,-1,-1,1)$. Their heads, with coordinates numbered $0, \dots, 23$ as in the octad tables, are

$$u_{\varepsilon,a} = 4\varepsilon e_1 + 2\big(a(e_2 - e_3) - e_4 + e_5 - \varepsilon a e_{13} + \varepsilon a(e_{19} + e_{22} + e_{23})\big), \qquad \varepsilon, a \in \{\pm 1\}.$$

Their squared norms as full rays are $96 \cdot 100 + 5 \cdot 31^2 = 14405$. The other 30 holes keep the weights just listed. The head-set assignment and the complete list of holes are specified by the accompanying coordinate file.

Finally add the two rational rays $Y_1 = (h_1, w_1)$ and $Y_2 = (h_2, w_2)$, where, using the same zero-based head coordinates,

$$\begin{aligned} h_1 &= 5e_4 - 6e_5 - 3e_{11}, & w_1 &= (-11, 23, 47, 34, -33), \\ h_2 &= (0,0,-1,-1,5,-4,-1,1,-1,0,0,0, & & \\ &\qquad -1,-2,0,0,0,0,1,0,0,-1,-1,-1), & w_2 &= (-41,-26,-27,-25,6). \end{aligned}$$

Their squared norms are 5174 and 3802. All rays, including the poles, are normalized individually.

Theorem 8.1. *This configuration is a kissing configuration of* $189168+21120+5+144+34+2 = 210473$ *points in* $\mathbb{R}^{29}$.

Proof. Pairs among the body and lifts are covered by Proposition 2.1. Layer–body pairs follow from Lemma 6.1(1). Against lifts, $\langle h, 4\sigma\rangle \le 16$ and $\langle t, w\rangle \le 1/\sqrt{2}$ give the same bound, less than 0.4928, as before. Between layer points, the argument of Theorem 7.2 applies: same directions have the same parity, opposite and orthogonal directions are harmless, and directions in $\mathcal{V}$ and $\mathcal{W}$ at inner product $\pm 1/2$ carry opposite parities.

For a hole ray $(p\sqrt{2}\,u, qs)$ of squared norm $N = 96p^2 + 5q^2$, the body product is at most $24\sqrt{2}\,p$ by (3). Thus the body condition follows from $N \ge 144p^2$. All five weight pairs satisfy this. For the four reweighted holes the difference is $14405 - 14400 = 5 > 0$. For Y_1 and Y_2, the head ℓ_1 norms are 14 and 21. Since every body coordinate has magnitude at most 4, their body products are at most 56 and 84, and

$$4 \cdot 56^2 < 32 \cdot 5174, \qquad 4 \cdot 84^2 < 32 \cdot 3802.$$

The remaining pair conditions involve the explicit poles, holes and rational-head points. They are checked directly by (1) in exact arithmetic on the final coordinate file, including every pair involving a reweighted hole, the fifth pole, or either rational-head point. Section 11

reports the full all-pairs check. For example, the product of each reweighted hole with Y_2 is $3875 - 180\sqrt{2}$, and the fifth pole has product 2198 with Y_2; both satisfy the criterion with the norms given above. Positive norms and the pair criterion also prove distinctness. □

9. Dimension 30

The configuration starts from a Leech lift: the 72 tails of type E_6 in 24 triangles, the 72 poles of [6], and 24 head sets on the octads of Table 5. It adds 72 hole points and 24 layer points.

9.1. **Hole points.** A unit vector $y \in \mathbb{R}^6$ is a *minuscule direction* if $\langle t, y\rangle \in \{0, \pm\frac{\sqrt{6}}{4}\}$ for every tail t; these are the directions of the 54 weights in the two minuscule orbits of E_6. The 72 hole points are $x(u, y)$, of the first kind of Section 6, for 72 pairs (u, y), with 36 distinct vectors $u \in \Lambda$ of squared norm 48, all of shape $(\pm 4, \pm 2^8, 0^{15})$, and 18 distinct minuscule directions y. The following facts were checked by computer:

(C5) for every hole point $x(u, y)$, every triangle not perpendicular to y carries a head set with $\langle u, h\rangle \le 16$ for all its heads h;
(C6) $\langle y, p\rangle \le \frac{1}{\sqrt{2}}$ for every y and every pole p;
(C7) the hole points have pairwise cosines at most $\frac{1}{2}$.

9.2. **Layer points.** Let $S' = \{3, 4, 17, 20\}$, a tetrad in O_{23}. Let w_1, w_2, w_3 be three unit vectors of the form $(t_i + t_j)/\sqrt{2}$ with $t_i \perp t_j$ tails, with $\langle w_a, w_b\rangle = -\frac{1}{2}$ for $a \ne b$. The 24 layer points are $Y(4\tau, w_a)$ with $\tau \in \{\pm 1\}^{S'}$ even and $a \in \{1, 2, 3\}$. A tail has inner product in $\{0, \pm\frac{1}{2\sqrt{2}}, \pm\frac{1}{\sqrt{2}}\}$ with each w_a. The following facts were checked by computer:

(C8) for each a, the 24 triangles split into 8 whose largest inner product with w_a is $\frac{1}{\sqrt{2}}$, 8 where it is $\frac{1}{2\sqrt{2}}$, and 8 perpendicular to w_a, and the split is the same for $a = 1, 2, 3$;
(C9) the head sets on the first 8 triangles have $\langle h, 4\tau\rangle \le 16$ and those on the next 8 have $\langle h, 4\tau\rangle \le 24$, for all heads h and all even τ;
(C10) $\langle w_a, p\rangle \in \{0, \pm\sqrt{3/8}\}$ for every pole p;
(C11) $\langle u, 4\tau\rangle \in \{\pm 8, \pm 24\}$ and $\langle w_a, y\rangle \in \{0, \pm\frac{\sqrt{3}}{4}\}$ for every hole point $x(u, y)$.

Among the head sets on O_{23}, those whose matchings are S'-bipartite satisfy (C9) with bound 16 by Lemma 4.4. The three sets on O_{23} whose matchings have a pair inside S' reach 32 and sit on perpendicular triangles. The other values in (C9) are a computer check.

Theorem 9.1. *This configuration is a kissing configuration of* $183888 + 38016 + 72 + 72 + 24 = 222072$ *points in* $\mathbb{R}^{30}$.

Proof. Pairs among the body, lifts and poles are covered by Proposition 2.1 and Section 3.

Hole points. Against the body, $\langle x(u, y), (\hat{b}, 0)\rangle = \alpha\frac{\langle u,b\rangle}{\sqrt{48\cdot 32}} \le \sqrt{\frac{2}{3}} \cdot \frac{24}{16\sqrt{6}} = \frac{1}{2}$ by (3). Against a lift, the value is $\frac{2}{3}\frac{\langle u,h\rangle}{16\sqrt{6}} + \frac{1}{3}\langle t, y\rangle$. On a triangle not perpendicular to y, (C5) gives at most $\frac{2}{3\sqrt{6}} + \frac{\sqrt{6}}{12} = \frac{7\sqrt{6}}{36} \approx 0.4763$. On a perpendicular triangle, $\langle u, h\rangle \le 24$ gives at most $\frac{1}{\sqrt{6}}$. Against a pole, the value is $\beta\langle y, p\rangle \le \frac{1}{\sqrt{6}}$ by (C6). Between hole points, (C7).

Layer points. Against the body, Lemma 6.1(1). Against lifts, (C8), (C9) and Lemma 6.1(2) show that the value is at most $\frac{16}{32\sqrt{6}} + \frac{1}{\sqrt{12}} \approx 0.4928$, $\frac{24}{32\sqrt{6}} + \frac{1}{2\sqrt{12}} \approx 0.4505$ and $\frac{32}{32\sqrt{6}} = \frac{1}{\sqrt{6}}$ on the three kinds of triangles. Against poles, (C10) gives at most $\frac{\sqrt{3}}{4}$. Against a hole point $x(u, y)$, the value is $\frac{\langle u,4\tau\rangle}{96} + \frac{\langle w_a,y\rangle}{\sqrt{6}}$, at most $\frac{1}{4} + \frac{\sqrt{2}}{8} \approx 0.4268$ by (C11). Between layer points

at the same w_a, distinct even τ give $\langle 4\tau, 4\tau'\rangle \le 0$; at different directions, the value is at most $\frac{64}{128} - \frac{1}{4} = \frac{1}{4}$. □

10. Dimension 31

The configuration starts from a Leech lift: the 126 tails of type E_7 in 42 triangles, the poles of [6], and 42 head sets on the octads of Table 5. The partition of the tails into triangles, the head sets and their assignment were chosen together by a constraint solver. One pole is removed and 40 layer points are added, in three classes:

- V_0: $Y(4\sigma, w)$ with σ odd on $\{10, 19, 21, 22\}$ and w one of three directions $w_0, w_\pm$ (24 points);
- X_1: $Y(4\sigma, w_0')$ with σ odd on $\{2, 5, 8, 11\}$ (8 points);
- X_3: $Y(4\sigma, w_3)$ with σ odd on $\{7, 8, 13, 14\}$ (8 points).

In the tail coordinates of the coordinate list,

$$w_0 = -e_4, \quad w_\pm = \tfrac{1}{543}(\pm 358, 0, 0, 166, 0, 0, \pm 373),$$
$$w_0' = e_4, \quad w_3 = \tfrac{1}{8\sqrt{2}}(4\sqrt{2}, 0, 4\sqrt{2}, -4\sqrt{2}, -4, 4, 0),$$

where $358^2 + 166^2 + 373^2 = 543^2$. The pairwise inner products of w_0, w_+, w_- are $-\frac{166}{543}$, $-\frac{166}{543}$ and $\frac{166^2 - 358^2 - 373^2}{543^2} \approx -0.813$, all negative, and $\langle w_0', w_3\rangle = -\frac{1}{2}$. The following facts were checked by computer:

(C12) every lift has cosine at most $\frac{1}{2}$ with every layer point; the largest value is about 0.4974;
(C13) every remaining pole p has $\langle w, p\rangle \le \frac{1}{\sqrt{2}}$ for all five directions w; the removed pole has inner product above $\frac{1}{\sqrt{2}}$ with w_3.

For (C12) Lemma 6.1(2) shows what is needed: triangles with a tail at inner product more than about 0.4747 with w must carry head sets with $\langle h, v\rangle \le 16$ for the layer vectors v of that class, and triangles with a tail at more than 0.2247 must carry sets with $\langle h, v\rangle \le 24$. For V_0 the tetrad $\{10, 19, 21, 22\}$ lies in O_{21}, and the head sets on O_{21} with $\{10, 19, 21, 22\}$-bipartite matchings have $\langle h, v\rangle \le 16$ by Lemma 4.4.

Theorem 10.1. *This configuration is a kissing configuration of* $174384 + 66528 + 125 + 24 + 8 + 8 = 241077$ *points in* $\mathbb{R}^{31}$.

Proof. Pairs among the body, lifts and poles are covered by Proposition 2.1 and Section 3; removing a pole does not hurt. Layer–body pairs are covered by Lemma 6.1(1), layer–lift pairs by (C12), and layer–pole pairs by (C13) and Lemma 6.1(3).

Layer, layer. By Lemma 6.1(4) the value is $\langle v, v'\rangle/128 + \langle w, w'\rangle/2$. Within one class at one direction, $\sigma \ne \sigma'$ have the same parity, so $\langle v, v'\rangle \le 0$. Within V_0 at different directions, $\langle w, w'\rangle < 0$ and $\langle v, v'\rangle \le 64$, so the value is below $\frac{1}{2}$. The tetrad of V_0 is disjoint from those of X_1 and X_3, so $\langle v, v'\rangle = 0$ across these classes and the value is at most $\frac{1}{2}$. The tetrads of X_1 and X_3 meet in one point, so $|\langle v, v'\rangle| \le 16$, and the value is at most $\frac{16}{128} - \frac{1}{4} < 0$. □

Theorems 7.2, 8.1, 9.1 and 10.1 prove Theorem 1.1.

11. Verification

What is proved by hand and what by computer. By hand: Proposition 2.1, Lemmas 2.2, 4.1, 4.2, 4.4, 4.5, 6.1 and 7.1, Theorem 4.3, and the case analyses in the proofs of Theorems 7.2 to 10.1. By computer, in exact arithmetic: the facts on tails and poles in Section 3, the subcodes of Table 4 and the cosets γ_i, the disjointness of the head sets, the values 16 in

TABLE 6. The exact all-pairs check.

dimension	points	pairs	result
28	205172	21 047 672 206	pass
29	210473	22 149 336 628	pass
30	222072	24 657 875 556	pass
31	241077	29 058 939 426	pass

Section 7, the final pole, hole and rational-head pair checks in Section 8, the facts (C5)–(C13), and the all-pairs check below.

Each configuration of Theorem 1.1 is given as a list of vectors whose coordinates are integer combinations of $1, \sqrt{2}, \sqrt{3}, \sqrt{6}$; the configuration consists of the normalized vectors. Both sides of (1) then lie in $\mathbb{Z}[\sqrt{2}, \sqrt{3}]$, and the sign of an element $a + b\sqrt{2} + c\sqrt{3} + d\sqrt{6}$ with integer a, b, c, d can be decided exactly by squaring twice.

All pairs. One program, written independently of the programs that built the configurations, checks each configuration on all pairs, with no symmetry reduction and no floating point in any decision. It confirms the number of points and that every point is nonzero. For every pair of distinct points x, y it decides (1), that is $\langle x, y\rangle \le \frac{1}{2}|x||y|$, by exact sign tests in $\mathbb{Z}[\sqrt{2}, \sqrt{3}]$, with 256-bit integers for the intermediate squares; an overflow would count as a failure. It also confirms that no two points have the same direction. Table 6 gives the results. Every pair passes, and in each configuration the largest cosine is exactly $\frac{1}{2}$. This check alone proves Theorem 1.1. The proofs by hand explain why the configurations work.

Structure. Separate programs recovered the body, head sets and lifts and checked the 528-point head-set structure, subcode cosets, disjointness and matchings described above. These components are unchanged by the final dimension-29 additions. The layer and hole heads lie in Λ; the two rational heads in dimension 29 are treated separately and are not assumed to be Leech vectors. Earlier scans of auxiliary cosines in floating point are explanatory diagnostics, not certificates for the final reweighted points. The decisive certificate for every final configuration is the exact all-pairs check on its exported coordinate file.

12. Conclusion

We have shown that $\tau_{28} \ge 205172$, $\tau_{29} \ge 210473$, $\tau_{30} \ge 222072$ and $\tau_{31} \ge 241077$, improving the best published lower bounds of 204522, 209497, 220440 and 238354.

In the Leech lift with the tails of [6], $\sum_i (|T_i| - 1)$ is already at the bound of Lemma 2.2. We do not know the largest head set. Each additional vector in every head set would add $\sum_i (|T_i| - 1)$ points, that is 16, 26, 48 and 84 in dimensions 28 to 31.

Layer and hole points replace some poles. The gain of layer points is limited by the head sets: a layer point needs head sets with small inner products against its vector v on the triangles near its direction. Lemma 4.4 gives such head sets on the octads through the tetrad of v, at most 20 of them by Lemma 4.5. In dimensions 30 and 31, where most head sets are elsewhere, there are few layer points. We do not know how far this can be pushed.

Use of artificial intelligence

GPT-6-Astra Pro and Claude Opus 5.5 were used to find these constructions and draft this paper.

Data availability

The coordinates and an exact checking program accompany this paper.

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